

Algorithmic Methods of Data Mining and Laboratory

Master's Degree in Data Science, Sapienza Università di Roma
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Reference Bibliography

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Disclaimer

This document contains notes for the **Algorithmic Methods of Data Mining and Laboratory** course, given by **Professors Anagnostopolous and Siciliano** as part of the Master's Degree in Data Science at **Sapienza Università di Roma** during the **first** semester of Academic Year **2026/2027**.

Note: These notes may not cover all details of each lecture, so it is highly suggested to consult the Professor's official resources and materials.

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Contacts

For further information or requests, feel free to contact me privately or reach out through my repository: <https://github.com/GianmariaRomano/Data-Science-Notes>.

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Chapter 1

Introduction to data mining

1.1 Finding a definition of data mining

While no unique definition exists yet, it is possible to view data mining as the use of efficient techniques for the analysis of very large collection of data and the extraction of useful and possibly unexpected patterns in data in order to make decisions across various areas.

Example: Example: Descriptive and predictive data mining

Descriptive data mining aims to identify patterns and relationships within data using unsupervised learning techniques that can be applied to recommendation systems, such as Netflix or Spotify's engines, to analyse patterns in user behaviour.

By contrast, predictive data mining focuses on using past observations to predict a target variable through supervised learning techniques.

For instance, CERN's *A Large Ion Collided Experiment* analyses heavy-ion collision data to investigate the properties of quark-gluon plasma.

1.2 Understanding data types

Since data can come in different forms, it is necessary to organize it in order to extract meaningful information or patterns.

In particular, data can be classified into three main types:

- **Structured data:** Data is organized according to well-enforced schema that is enforced during storage.
- **Semi-structured data:** Data can be organized according to a structural schema, although the organization is not enforced during storage.
- **Unstructured data:** Data does not follow a pre-defined structure, requiring a data cleaning procedure to store it as (semi-)structured data before use.

Example: Example: Different data types

Examples of data organization can be seen on the Infostud platform, which enforces strict organization through dedicated constraints on certain types of data, and medical records, which follow some structural patterns but might contain errors or anomalous data.

By contrast, social media platforms, such as Twitter or Reddit, provide clear examples of unstructured data that do not follow an organizational structure.